Using scripts to create site content (JS)

Learning objectives:

- use various algorithmic structures in script language
- use script language in site design
- use script language to provide interactivity

Using scripts to create site content

Assessment Criteria:

- Use various algorithmic structures in script language
- Use script language in site design
- Use script language to provide interactivity

What Can JavaScript Do?

- JavaScript is the programming language of the web.
- It can calculate, manipulate and validate data.
- It can update and change both HTML and CSS.

JavaScript can change HTML content

```
<!DOCTYPE html>
<html>
<body>

<p id=“test">JavaScript can change HTML content.</p>

<button type="button" onclick='document.getElementById(" test
").innerHTML = "Hello JavaScript!"'>Click Me!</button>

</body>
</html>
```

JavaScript can change HTML attributes

```
<!DOCTYPE html>
<html>
<body>
<button onclick="document.getElementById('myImage').src='pic_bulbon.gif'">Turn on the light</button>

<img id="myImage" src="pic_bulboff.gif" style="width:100px">

<button onclick="document.getElementById('myImage').src='pic_bulboff.gif'">Turn off the light</button>

</body>
</html>
```

JavaScript can change CSS style

```
<!DOCTYPE html>
<html>
<body>
<button type="button"
onclick="document.getElementById('demo').style.fontSize='35px'">Click
Me!</button>

</body>
</html>
```

Where to Insert JavaScript

- JavaScript in <head>
- JavaScript in <body>
- JavaScript in an external file

JavaScript in <head>

```
<!DOCTYPE html>
<html>
<head>
<script>
function myFunction() {
  document.getElementById("demo").innerHTML = "Paragraph changed.";
}
</script>
</head>
<body>
<p id="demo">A Paragraph.</p>
<button type="button" onclick="myFunction()">Try it</button>
</body>
</html>
```

JavaScript in <body>

```
<!DOCTYPE html>
<html>
<body>
<p id="demo">A Paragraph.</p>
<button type="button" onclick="myFunction()">Try it</button>
<script>
function myFunction() {
  document.getElementById("demo").innerHTML = "Paragraph changed.";
}
</script>
</body>
</html>
```

JavaScript in an external file

```
<!DOCTYPE html>
<html>
<body>
<p id="demo">A Paragraph.</p>
<button type="button" onclick="myFunction()">Try it</button>
<p>This example links to "myScript.js".</p>
<p>(myFunction is stored in "myScript.js")</p>
<script src="myScript.js"></script>
</body>
</html>
```

JavaScript Syntax

```
// How to Declare variables:
let x = 5;
let y = 6;

// How to Compute values:
let z = x + y;

// I am a Comment. I do Nothing
```

If else

- ```
  if (condition1) {
    // code to execute if condition1 is true
  } else if (condition2) {
    // code to execute if the condition1 is false and condition2
  is true
  } else {
    // code to execute if the condition1 is false and condition2
  is false
  }
  ```
  ```

Loop for

- ```
  for (expr1; expr2; expr) {
    // code block to be executed
  }
  ```

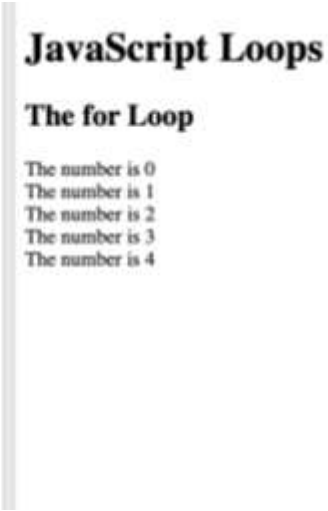
```
<!DOCTYPE html>
<html>
<body>
<h1>JavaScript Loops</h1>
<h2>The for Loop</h2>

<script>
let text = "";

for (let i = 0; i < 5; i++) {
  text += "The number is " + i + "<br>";
}

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```


```

JS functions

```
<!DOCTYPE html>
<html>
<body>
<h1>JavaScript Functions</h1>
<h2>The return Statement</h2>
<p>A function that returns the greeting: "Hello World":</p>

<p id="demo"></p>

<script>
function sayHello() {
  return "Hello World";
}

let greeting = sayHello();

document.getElementById("demo").innerHTML = greeting;
</script>

</body>
</html>
```



JavaScript Events

- HTML events are **things that happen** to HTML elements.

- **Examples of events:**
- An HTML button is clicked
- A web page has finished loading
- The mouse moves over an element
- A keyboard key is pressed
- An HTML input field is changed

Example of events

```
<!DOCTYPE html>
<html>
<body>
<h1>JavaScript HTML Events</h1>
<h2>The onclick Attribute</h2>

<button
onclick="document.getElementById('demo').innerHTML=Date()">The
time is?</button>

<p id="demo"></p>

</body>
</html>
```



Common HTML events

Event	Description
onchange	An HTML element has been changed
onclick	The user clicks an HTML element
onmouseover	The user moves the mouse over an HTML element
onmouseout	The user moves the mouse away from an HTML element
onkeydown	The user pushes a keyboard key
onload	The browser has finished loading the page

Read JS syntax in https://www.w3schools.com/js

Practical task. Write JS code for the project To-Do list

- use script language in site design
- use script language to provide interactivity

```
<!DOCTYPE html>
<html>
<body>
<h1>JavaScript Project</h1>
<h2>To-Do List</h2>

<input id="task" placeholder="New task">

<button onclick="addTask()">Add</button>

<ul id="list"></ul>

<button onclick="clearAll()">Clear All</button>

<script>
// Create a task Array
let tasks = [];
</script>

</body>
</html>
```



Task-2. Working with Forms

submit form

```
<!DOCTYPE html>
<html>
<body>
<form id="frm1" action="/action_page.php">
  First name: <input type="text" name="fname"><br>
  Last name: <input type="text" name="lname"><br><br>
  <input type="button" onclick="myFunction()"
value="Submit">
</form>
<script>
function myFunction() {
  document.getElementById("frm1").submit();
}
</script>
</body>
</html>
```

reset form

```
<!DOCTYPE html>
<html>
<body>
<form id="frm1">
  First name: <input type="text" name="fname"><br>
  Last name: <input type="text" name="lname"><br><br>
  <input type="button" onclick="myFunction()" value="Reset">
</form>
<script>
function myFunction() {
  document.getElementById("frm1").reset();
}
</script>
</body>
</html>
```

